Current: charge flow rate

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Learning target: Calculate current as charge transferred per time.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 2.*
Force magnitude becomes 3 times as large, from Coulomb’s law.

02. Worked example: explain the model
Same steady series current through each section; energy is transferred, charge not used up.

03. Read and interpret the model
Conventional current follows that loop direction; steady current stops when the path opens.

04. Apply the physics
I = 12/7 ≈ 1.71 A. A constant-current graph is a straight line from (0 s, 0 C) to (7 s, 12 C).

05. Explain your reasoning
3 A; charge does not steadily accumulate at the junction.

06. Review: connect an earlier idea *Review: Unit 6, Day 11.*
Observed frequency is higher than 600 Hz because arriving wavefronts are closer together in time.

07. Exit check
6 C.

08. STRETCH: test the model
Examples 4 C/2 s and 10 C/5 s; current is rate.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 2.*
$F_f/F_i=9/25\approx0.36$.

02. Worked example: explain the model
Same steady series current through each section; energy is transferred, charge not used up.

03. Read and interpret the model
Conventional current follows that loop direction; steady current stops when the path opens.

04. Apply the physics
$q=It=0.25\times480=120$ C.

05. Explain your reasoning
3 A; charge does not steadily accumulate at the junction.

06. Review: connect an earlier idea *Review: Unit 6, Day 11.*
$f_{obs}\approx637.5$ Hz, higher than emitted frequency.

07. Exit check
No; the circuit’s electromagnetic response propagates through the circuit while local charges drift much more slowly.

08. STRETCH: test the model
Triangle 6 C plus rectangle 8 C=14 C.